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THEORY OF CRYSTAL STRUCTURES

Combinatorial Approach to the Simulation of Superstructures of the NaCl Type

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Abstract—The combinatorial approach is used to simulate superstructures of the NaCl type. An algorithm for searching for possible models of superstructures is developed which includes alternating the occupation of the metal sublattice by chemically different cations in an isolated cubic fragment. The application of this algorithm to small cubic fragments is demonstrated. The models are compared with known complex metal oxides (binary *A*–*B*–O and ternary *A*–*B*–*C*–O compounds).

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INTRODUCTION

The NaCl structure type (halite) is widespread among inorganic compounds (it includes more than 200 representatives) [1]. In addition, 4599 compounds are assigned to this type in modern databases (for example, Pearson's Crystal Data). Many compounds with *AX* stoichiometry—most alkali metal and silver halides, chalcogenides of magnesium and alkaline earth metals, some nitrides and phosphides, interstitial phases (oxides, nitrides, carbides), ionic–covalent compounds of transition metals, and a number of intermetallides—are crystallized into this structure. The reason for the large variety of compounds and types of chemical bonds in them is that this structure allows for not only *A*–*X*, but also *A*–*A* and *X*–*X* bonds and, in addition, can be implemented in a wide range of ratios of *A* and *X* radii [1]: $0.44 < r_A/r_X < 1.52$ (from 0.44 (LiI) to 1.52 (CsF), according to the PDF-2 data and the system of Shannon crystalline radii [2]).

The NaCl structure is a typical representative of the sp. gr. $Fm\bar{3}m$ (no. 225) with a maximum number of possible symmetry elements. In this structure, *A* and *X* atoms alternate in a simple cubic packing. This is the only *AX* structure with a regular octahedral coordination of atoms of both types [3]. A wide review and systematization of superstructures, starting from the halite type, was presented in [4] with allowance for the valence strengths and ratios of ionic radii.

Global and Local Valence Balances in Halites

The global valence balance in halites indicates electroneutrality (the sum of cationic and anionic charges is zero). We will consider stoichiometric compounds $A_aB_bX_x$ as an example. Let *A* and *X* be a singly charged cation and an oxygen anion, respectively. Then, for *B*[+]-cation oxidation states of +3, +4, +5, +6, and +7, we obtain stoichiometric halites ABO_2, A_2BO_3, A_3BO_4, A_4BO_5, and A_5BO_6, respectively.

The local valence balance in halites was calculated according to [5] using the formula for cation valence strength: $v = z/n$ (*z* is the charge and *n* is the coordination number). According to the second Pauling rule [6], the sum of the valence strengths of the six cations that coordinate oxygen should be approximately equal (in modulus) to the oxygen charge.

In this case the local valence balance can be implemented only at *B* + oxidation states of +3, +4, and +7:

$$1/2A^{+1}B^{+3}O_2 = A^{+1}_{3/6}B^{+3}_{3/6}O,$$
$$\sum z/n = 3 \times 1/6 + 3 \times 3/6 = 2;$$
$$1/3A^{+1}_2B^{+4}O_3 = A^{+1}_{4/6}B^{+4}_{2/6}O,$$
$$\sum z/n = 4 \times 1/6 + 2 \times 4/6 = 2;$$
$$1/6A^{+1}_5B^{+7}O_6 = A^{+1}_{5/6}B^{+7}_{1/6}O,$$
$$\sum z/n = 5 \times 1/6 + 1 \times 7/6 = 2.$$

In this case there are 3, 2, and 1 highly charged neighbors in the oxygen environment, respectively.

This approach is valid only for stoichiometric compounds AMO_2, A_2MO_3, and A_5MO_6; i.e., it contradicts the experimental data. Indeed, well-known halites such as Li_3TaO_4, Na_3SbO_4, and Li_4UO_5 [4] turn out to be forbidden. The possibility of distorting octahedra must also be taken into account. The dependence that the valence strength has on distance is known. One example is three five-charged cations around oxygen in Li_3NbO_4 [4]: the sum of charges in the first coordination sphere of oxygen (octahedron) is 18 (three Li^{+1} and three Nb^{+5} ions) rather than 12 at one of them and 10 (five Li^{+1} and one Nb^{+5} ions) rather than 12 at each of the other three. However, a simulation based on

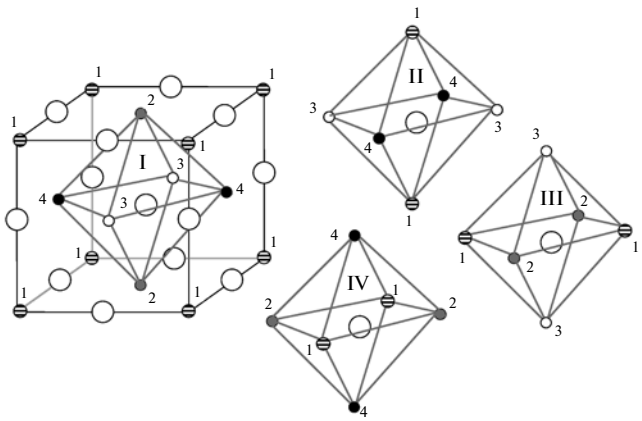

Fig. 1. Coloring of CF4 in four colors; octahedra I, II, III, and IV.

ideal cells (the geometric approach) inevitably leads back to Pauling's old formula that was applied by Hauck. In this study we propose taking into *account distortions as a factor leading to nonequivalent valence strengths*.

Eremin [7] performed a systematic analysis of the method of atomistic simulation of the crystal structure and properties of minerals and inorganic compounds, showed its advantages in comparison with other methods for predicting atomic structure and properties of crystals, and indicated its applicability limits. However, models of superstructures of the NaCl type in supercells of a specified size, with allowance for all stoichiometries and all possible chemically different cations, were not completely considered in [7].

1. STATEMENT OF THE PROBLEM

To reveal the structural possibility of implementing superstructures of the NaCl type, we used a combinatorial analysis (combinations; permutations; and arrangements of cations with different charges, taking into account the global valence balance) in a cubic region, which can be a translational unity in the lattice (the smallest of them coincides with the NaCl unit cell). The input enumeration parameters are as follows: composition and stoichiometry from $A_aB_b\mathrm{O}^{2-}_{(a+b)}$ to $A_aB_bC_cD_dE_eI_iF_f\mathrm{O}^{2-}_{(a+b+c+d+e+i+f)}$, where A, B, C, D, E, I, and F are cations with different charges (from +1 to +7) and a, b, c, d, e, i, and f are integers. This problem cannot be solved by simple enumeration (in which case even the supercomputer capacity is insufficient); therefore, we considered a more realistic algorithm of combinatorial analysis with the aim of obtaining models of superstructures of certain chemical compositions and stoichiometry based on an analysis of small cubic fragments.

2. ALGORITHM OF COMBINATORIAL ANALYSIS

In superstructures of the NaCl type, cation and/or anion sublattices are inhomogeneous: each of them contains two or more crystallographically different types of atoms (which are arranged regularly). Furthermore we will be interested in only the charge filling of the metal sublattice.

To solve this problem, it is necessary to enumerate all versions of the mutual arrangement of all possible charges (from +1 to +7) in the metal sublattice. In the first stage, we take into consideration only the mutual arrangement of different cations and disregard the valence balance. Therefore, it is convenient to introduce a concept like coloring, i.e., marking a conditional charge by a certain color (since the charges are unknown, the color value is of no importance; it is their difference that matters). Then the simulation of superstructures is reduced to an enumeration of all possible arrangements of colors in the cubic fragment (see description below).

In the next stage, it is necessary to choose certain charges for each color in the cubic fragment (taking into account the global and local valence balances). As a result we obtain models of superstructures having a specific structure, stoichiometry, and charge distribution. Cubic fragments can also be systematized according to symmetry elements.

2.1. Choice of Cubic Fragment

To generate various superstructures, we choose a cubic fragment (CF) satisfying the following requirements:

(i) The smallest CF coincides with the NaCl unit cell and can be used as a reference point.

(ii) At certain external coloring, CF has a translational periodicity (opposite faces have the same color); i.e., the symmetry of the metric is at least orthorhombic. On the whole, cells can be cubic, tetragonal, monoclinic, rhombohedral, triclinic, or hexagonal. A hexagonal structure can be obtained from CF by the contracting of extending it along a threefold axis.

(iii) Depending on the external and internal coloring, CF may either coincide with the unit cell or contain several unit cells.

(iv) NaCl positions are associated with the charge, which affects the valence balance, thus greatly facilitating the search for superstructures. Therefore, charges of the same type in the CFs can be differently colored, imitating different crystallographic types of atoms (the addition of crystallographic types may reduce symmetry).

To enumerate all versions of filling the cation sublattice, CF sizes (the number of ions per edge is odd) and the number of different cations (with different charges) are specified. Then the cation sublattice is alternately filled by different cations. As an example, Fig. 1 shows CF coloring in four colors. This technique seems to be difficult to implement because of a very large number of enumeration versions. In particular, the number of cations for CF with three ions per edge is 14; correspondingly, the number of versions is 7^{14} (7 is the number of cations with charges from +1 to +7). However, Fig. 1 (hereinafter, small and large balls in figures indicate metal cations and O^{2-} anions, respectively) demonstrates that some cation positions are equivalent, because they are located on the surface of a CF possessing translational periodicity. Therefore, the problem is significantly simplified and the maximum number of different cation sublattices is reduced to 7^4.

2.2. CF Sizes and Properties

The CF size is determined from the formula

$$n = 4 \times x^3,$$

where n is the number of formula units (FUs) in the CF under consideration, 4 is the number of FUs in the minimum CF, and x is the number of shortest cubic translations.

The number of independent positions in the CF is determined by the number of cations in the CF and on the surface, with allowance for the mirror coloring of opposite faces.

Let us consider the following examples. An octahedral coordination sphere yields 6 independent positions (IPs) (if all vertices are located on the CF surface, there are 3 IPs), a cubic coordination sphere yields 8 IPs (with all vertices on the CF surface, there is only 1 IP), and a 24-vertex polyhedron (the next coordination sphere) yields 24 IPs (with all vertices on the CF surface, there are 12 IPs). A CF with four FUs consists of an octahedron and a cube, the vertices of which lie on the surface; correspondingly, there are 3 + 1 = 4 IPs of cations.

It is convenient to separate CFs according to the chemical composition of a cation sublattice (different charges) from 2 to 7 (one color does not imply coloring versions). For convenience, we will correlate charge with color (colors from I to VII correspond to charges from +1 to +7, respectively). The maximally possible number of colors is seven, but no larger than the number of IPs in CF. In a CF with four FUs, the maximum number of colors is four.

CF sizes and characteristics (any coloring inside cube); the number of IPs for each coordination sphere is given in parentheses:

$4 \times 1^3 = 4$ FUs, 8 atoms, (3 + 1) = 4 IPs;

Table 1. Sixteen formal versions of CF4 coloring

Version	Positions			
	1	2	3	4
1	0	0	0	0
2	0	0	0	1
3	0	0	1	0
4	0	0	1	1
5	0	1	0	0
6	0	1	0	1
7	0	1	1	0
8	0	1	1	1
9	1	0	0	0
10	1	0	0	1
11	1	0	1	0
12	1	0	1	1
13	1	1	0	0
14	1	1	0	1
15	1	1	1	0
16	1	1	1	1

Table 2. Two groups of coloring with different color ratios

Version	3 : 1				Version	1 : 1			
	1	2	3	4		1	2	3	4
2	0	0	0	1	4	0	0	1	1
3	0	0	1	0	6	0	1	0	1
5	0	1	0	0	7	0	1	1	0
9	1	0	0	0	10	1	0	0	1
					11	1	0	1	0
					13	1	1	0	0

$4 \times 2^3 = 32$ FUs, 64 atoms, ((6 + 8) + 12 + 6) = 32 IPs;

$4 \times 3^3 = 108$ FUs, 216 atoms, ((6 + 8 + 24 + 27) + 12 + 12 + 12 + 6 + 1) = 108 IPs;

$4 \times 4^3 = 256$ FUs, 512 atoms, 256 IPs;

$4 \times 5^3 = 500$ FUs, 1000 atoms, 500 IPs;

$4 \times 6^3 = 864$ FUs, 1728 atoms, 864 IPs;

$4 \times 7^3 = 1372$ FUs, 2744 atoms, 1372 IPs.

3. APPLICATION OF THE ALGORITHM OF COMBINATORIAL ANALYSIS

3.1. Enumeration of Colorings for CFs with Four FUs

Figure 1 shows the only coloring of a CF with four FUs (CF4) in four colors (the positions of independent cations are enumerated).

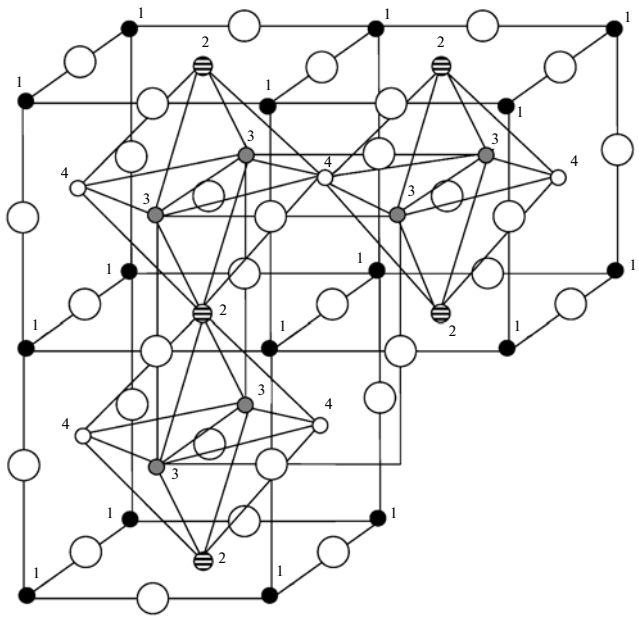

Fig. 2. Four interpenetrating CF4s.

I. For two types of cations there are formally $2^4 = 16$ versions. It is convenient to enumerate all coloring versions in the binary number system (conditional colors are, respectively, "0" and "1") from 0000 to 1111 (Table 1). However, after rejecting monochromatic versions and imposing a limitation on the contributions of each color (the sum of color "0" is no less than the sum of color "1"), only 10 versions remain. For convenience, we will divide them into groups (according to stoichiometry) (Table 2).

These groups are likely to contain identical CFs. Geometrically compatible CFs are obtained via rotation along the two-, three-, or fourfold axes. Interpenetrating CFs can describe the same structure, although colorings can be different (Fig. 2 shows that they differ by coloring vertices). Furthermore, we will use the so-called rejection algorithm.

3.1.1. Rejection algorithm. It can be seen in Fig. 2 that there may be four interpenetrating CFs (differing in vertex color), and each of them is represented by six geometrically compatible CFs. Thus, the total number of CFs describing one structure is 24 (Table 3).

Now, when comparing two CFs, one can determine if they belong to the same structure or to different structures. To this end, it is sufficient to substitute the coloring of one of them into the other 24 CFs (Table 3) and compare the sequence of color orders. For example, let us find out if CF 2 and CF 3 (from Table 2) belong to the same structure or to different ones. To this end, we substitute the colors of CF 3 (initial colors 0, 0, 1, and 0 correspond to positions 1, 2, 3, and 4) into all positions of Table 3 and compare with the alternation of colors in CF 2 (0, 0, 0, 1). Table 3 contains a CF with positions (1, 2, 4, 3); if we substitute the colors of CF 3 into it, its color sequence will coincide with that of CF 2 (0001), and CF 3 will be rejected.

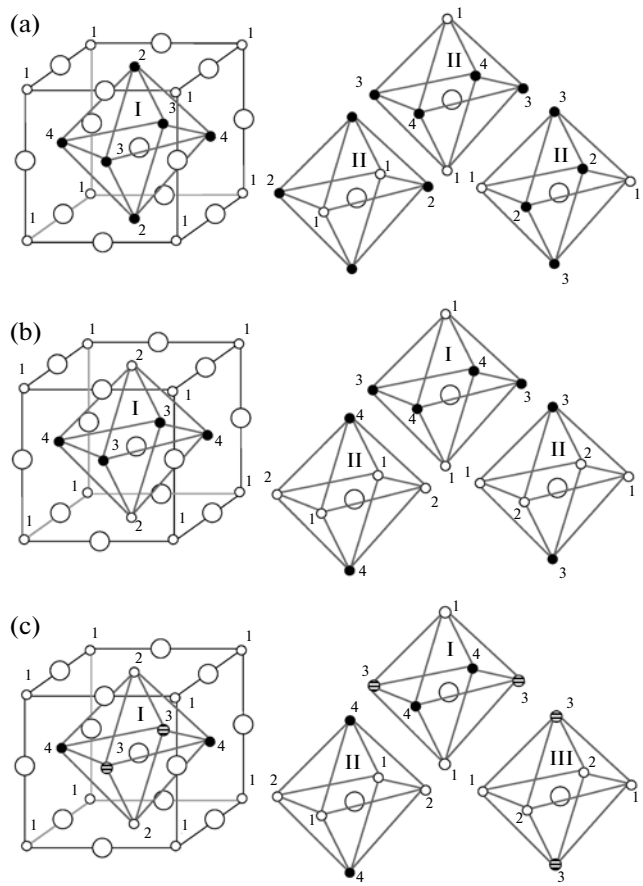

Fig. 3. Coloring versions after applying the rejection algorithm: (a) ratio 1 : 3, one octahedron of type I and three octahedra of type II; (b) ratio 1 : 1, two octahedra of type I and two octahedra of type II; and (c) ratio 1 : 1: 2, two octahedra of type I, one octahedron of type II, and one octahedron of type III.

After applying the rejection algorithm to all coloring versions (Table 2), only two different versions remain: version 2 (color ratio 3 : 1 (Fig. 3a)) and version 4 (color ratio 1 : 1 (Fig. 3b)).

II. In the case of three types of cations, there are formally $3^4 = 81$ versions. It is convenient to enumerate all coloring versions in the ternary number system (conditional colors are, correspondingly, "0," "1," and "2") from 0000 to 2222. However, after rejecting monochromatic and two-colored versions and introducing a limitation on the contributions of each color (sum of color "0" ≥ sum of color "1," sum of color "1" ≥ sum of color "2"), only 12 versions remain (Table 4).

After rejecting geometrically compatible and interpenetrating colorings (according to the rejection algo-

Table 3. Positions denoting geometrically compatible and interpenetrating CFs according to Fig. 2

CF	Positions
	Basic CF
1	1 4 2 3 (basic CF (axis passes through position 4)))
2	1 4 3 2 (geometrically compatible, rotation by 180° (axis passes through position 4)))
3	1 3 2 4 (geometrically compatible (axis passes through position 3))
4	1 3 4 2 (geometrically compatible, rotation by 180°)
5	1 2 4 3 (geometrically compatible (axis passes through position 2))
6	1 2 3 4 (geometrically compatible, rotation by 180°)
	1 interpenetrating CF
7	3 2 4 1 (geometrically compatible (axis passes through position 2))
8	3 2 1 4 (geometrically compatible, rotation by 180°)
9	3 4 2 1 (geometrically compatible (axis passes through position 4))
10	3 4 1 2 (geometrically compatible, rotation by 180°)
11	3 1 2 4 (geometrically compatible (axis passes through position 1))
12	3 1 4 2 (geometrically compatible, rotation by 180°)
	2 interpenetrating CF
13	4 1 2 3 (geometrically compatible (axis passes through position 1))
14	4 1 3 2 (geometrically compatible, rotation by 180°)
15	4 2 1 3 (geometrically compatible (axis passes through position 2))
16	4 2 3 1 (geometrically compatible, rotation bya 180°)
17	4 3 1 2 (geometrically compatible (axis passes through position 3))
18	4 3 2 1 (geometrically compatible, rotation by 180°)
	3 interpenetrating CF
19	2 1 3 4 (geometrically compatible (axis passes through position 1))
20	2 1 4 3 (geometrically compatible, rotation by 180°)
21	2 3 1 4 (geometrically compatible (axis passes through position 3))
22	2 3 4 1 (geometrically compatible, rotation by 180°)
23	2 4 1 3 (geometrically compatible (axis passes through position 4))
24	2 4 3 1 (geometrically compatible, rotation by 180°)

rithm conditions), only one stoichiometric version remains: 2 : 1: 1 (Fig. 3c).

III. If there are four types of cations, there are formally $4^4 = 256$ versions. It is convenient to enumerate all coloring versions in the quaternary system (conditional colors are, correspondingly, "0," "1," "2" and "3") from 0000 to 3333. However, after rejecting monochromatic, two-colored, and three-colored versions and introducing limitations on the contributions from each color (sum of color "0" ≥ sum of color "1," sum of color "1" ≥ sum of color "2," sum of color "2" ≥ sum of color "3"), only 24 versions remain (Table 5).

After rejecting geometrically compatible and interpenetrating colorings (according to the rejection algorithm conditions), only one stoichiometric version remains: 1 : 1: 1: 1 (Fig. 1).

Table 4. Intermediate stage of rejecting versions of CF4 coloring in three colors

Version	2 : 1 : 1
	1234
6	0012
8	0021
10	0102
16	0120
20	0201
22	0210
30	1002
34	1020
46	1200
56	2001
58	2010
64	2100

Table 5. Intermediate stage of rejecting versions of CF4 coloring in four colors

Version	1 : 1 : 1 : 1	Version	1 : 1 : 1 : 1	Version	1 : 1 : 1 : 1
	1234		1234		1234
28	0123	100	1203	178	2301
31	0132	109	1230	181	2310
40	0213	115	1302	199	3012
46	0231	121	1320	202	3021
55	0312	136	2013	211	3102
58	0321	142	2031	217	3120
76	1023	148	2103	226	3201
79	1032	157	2130	229	3210

Table 6. Versions of charge coloring of octahedron with local valence balance

Cations + 1, + 2, + 3, + 4, + 5, + 6, + 7
1 1 1 1 2 6
1 1 1 1 3 5
1 1 1 1 4 4
1 1 1 2 2 5
1 1 1 2 3 4
1 1 1 3 3 3
1 1 2 2 2 4
1 1 2 2 3 3
1 2 2 2 2 3
2 2 2 2 2 2
1 1 1 1 1 7

Table 7. CF32 (two different cations), possible stoichiometries

Ratio		Formula
31 : 1		no
30 : 2	15 : 1	no
29 : 3		no
28 : 4	7 : 1	no
27 : 5		no
26 : 6	13 : 3	no
25 : 7		no
24 : 8	3 : 1	no
23 : 9		no
22 : 10	11 : 5	no
21 : 11		no
20 : 12	5 : 3	no
19 : 13		no
18 : 14	9 : 7	no
17 : 15		no
16 : 16	1 : 1	$A^{+1}B^{+3}X_2^{-2}$

3.2. Applying Global Valence Balance to the Remaining Colorings

All possible versions of CF4 coloring, with allowance for the rejection of identical (geometrically compatible and interpenetrating) ones, were considered above. Now CF can be brought closer to the real model of the superstructure by assigning a certain charge to each color.

The sum of the anion and cation charges must be zero or the sum of cation charges for CF4 (in the case of doubly charged anion) is eight, while the cation charges may change from +1 to +7.

I. For the stoichiometry 1 : 3, the sum of cation charges $(a + 3 \times b) = 8$. It is convenient to enumerate all a and b in the septenary system from 12 to 76 (a and b are the first and second decimals, respectively). The global valence balance is satisfied only at the charges $a = 5$ and $b = 1$ ($A^{+5}B_3^{+1}X_4^{-2}$).

II. For the stoichiometry 1 : 1, the sum of cation charges $2 \times (a + b) = 8$. In this case, it is also convenient to enumerate all a and b in the septenary system from 12 to 76. The global valence balance is satisfied only at the charges $a = 1$ and $b = 3$ ($A^{+1}B^{+3}X_2^{-2}$).

III. For the stoichiometry 2 : 1: 1, the sum of cation charges $2 \times a + b + c = 8$. Similarly, it is convenient to enumerate all a, b, and c in the septenary system from 123 to 765 (the first, second, and third decimals are a, b, and c, respectively). The global valence balance is satisfied only at the following charge combinations: $a = 1$, $b = 2$, and $c = 4$ ($A_2^{+1}B^{+2}C^{+4}X_4^{-2}$) and $a = 2$, $b = 1$, and $c = 3$ ($A_2^{+2}B^{+1}C^{+3}X_4^{-2}$).

IV. For the stoichiometry 1 : 1: 1: 1, the sum of cation charges is $a + b + c + d = 8$. It is convenient to enumerate all a, b, c, and d in the septenary system from 1234 to 7654. The global valence balance is not satisfied for any charges.

Note that the local valence balance (Table 6) is not satisfied for any CF4 colorings.

3.3. Complete Analysis of a Cubic Fragment with 32 FUs

Table 7 contains all formal CF stoichiometries with 32 FUs (CF32) of two colors. However, Table 6 indicates that dichromatic coloring suggests the presence of only one type of octahedra: 111333, 111117, or 111144 (with allowance for the local valence balance); i.e., the CF32 stoichiometry should correspond to the composition of the octahedron coordinating the oxygen anion. Only one ratio is satisfied: 1 : 1 or $A^{+1}B^{+3}X_2^{-2}$.

A complete analysis of CF32 cannot be performed as easily as in the case of CF4, because now the number of independent positions is many times larger; the total number of versions is 7^{32} (sevenfold composition, 32IP). To solve this problem, a BERNSIDE program

Table 8. Two-component and three-component stoichiometries in CF32

Two-component	Three-component		
1. $A^{+1}B^{+3}X_2^{-2}$	19. $A_{17}^{+2}\ B_{10}^{+1}\ C_5^{+4}\ X_{32}^{-2}$	39. $A_{15}^{+1}\ B_{14}^{+2}\ C_3^{+7}\ X_{32}^{-2}$	58. $A_{22}^{+1}\ B_7^{+3}\ C^{+7}{}_3X_{32}^{-2}$
2. $A_3^{+1}\ B^{+5}X_4^{-2}$	20. $A_{18}^{+1}\ B_9^{+4}\ C_5^{+2}\ X_{32}^{-2}$	40. $A_5^{+1}\ B_2^{+2}\ C^{+7}X_8^{-2}$	59. $A_6^{+1}\ B^{+3}C^{+7}X_8^{-2}$
Three-component	21. $A_5^{+2}\ B_2^{+1}\ C^{+4}X_8^{-2}$	41. $A_{10}^{+2}\ B_5^{+1}\ C^{+7}X_{16}^{-2}$	60. $A_{26}^{+1}\ B_5^{+7}\ C^{+3}X_{32}^{-2}$
1. $A_{11}^{+1}\ B_{11}^{+3}\ C_{10}^{+2}\ X_{32}^{-2}$	22. $A_{10}^{+1}\ B_5^{+4}\ C^{+2}X_{16}^{-2}$	42. $A_{25}^{+1}\ B_5^{+7}\ C_2^{+2}\ X_{32}^{-2}$	
2. $A_6^{+2}\ B_5^{+1}\ C_5^{+3}\ X_{16}^{-2}$	23. $A_{23}^{+2}\ B_6^{+1}\ C_3^{+4}\ X_{32}^{-2}$	43. $A_{26}^{+2}\ B_5^{+1}\ C^{+7}X_{32}^{-2}$	61. $A_{11}^{+1}\ B_4^{+4}\ C^{+5}X_{16}^{-2}$
3. $A_3^{+1}\ B_3^{+3}\ C_2^{+2}\ X_8^{-2}$	24. $A_{23}^{+2}\ B_2^{+1}\ C^{+4}X_{16}^{-2}$		62. $A_{23}^{+1}\ B_5^{+5}\ C_4^{+4}\ X_{32}^{-2}$
4. $A_{13}^{+1}\ B_{13}^{+3}\ C_6^{+2}\ X_{32}^{-2}$	25. $A_{29}^{+2}\ B_2^{+1}\ C^{+4}X_{32}^{-2}$	44. $A_{17}^{+1}\ B_{13}^{+3}\ C_2^{+4}\ X_{32}^{-2}$	
5. $A_{14}^{+2}\ B_9^{+1}\ C_9^{+3}\ X_{32}^{-2}$		45. $A_9^{+1}\ B_5^{+3}\ C_2^{+4}\ X_{16}^{-2}$	63. $A_{22}^{+1}\ B_9^{+4}\ C^{+6}X_{32}^{-2}$
6. $A_7^{+1}\ B_7^{+3}\ C_2^{+2}\ X_{16}^{-2}$	26. $A_{15}^{+1}\ B_{12}^{+2}\ C_5^{+5}\ X_{32}^{-2}$	46. $A_{19}^{+1}\ B_7^{+3}\ C_6^{+4}\ X_{32}^{-2}$	64. $A_6^{+1}\ B^{+4}C^{+6}X_8^{-2}$
7. $A_{15}^{+1}\ B_{15}^{+3}\ C_2^{+2}\ X_{32}^{-2}$	27. $A_4^{+2}\ B_3^{+1}\ C^{+5}X_8^{-2}$	47. $A_5^{+1}\ B_2^{+4}\ C^{+3}X_8^{-2}$	
8. $A_2^{+2}\ B^{+1}C^{+3}X_4^{-2}$	28. $A_9^{+1}\ B_4^{+2}\ C_3^{+5}\ X_{16}^{-2}$	48. $A_{21}^{+1}\ B_{10}^{+4}\ C^{+3}X_{32}^{-2}$	65. $A_{19}^{+1}\ B_{11}^{+3}\ C_2^{+6}\ X_{32}^{-2}$
9. $A_{18}^{+2}\ B_7^{+1}\ C_7^{+3}\ X_{32}^{-2}$	29. $A_{20}^{+2}\ B_9^{+1}\ C_3^{+5}\ X_{32}^{-2}$		66. $A_{11}^{+1}\ B_3^{+3}\ C_2^{+6}\ X_{16}^{-2}$
10. $A_{10}^{+2}\ B_3^{+1}\ C_3^{+3}\ X_{16}^{-2}$	30. $A_{21}^{+1}\ B_7^{+5}\ C_4^{+2}\ X_{32}^{-2}$	49. $A_{17}^{+1}\ B_{14}^{+3}\ C^{+5}X_{32}^{-2}$	67. $A_{25}^{+1}\ B_6^{+6}\ C^{+3}X_{32}^{-2}$
11. $A_{22}^{+2}\ B_5^{+1}\ C_5^{+3}\ X_{32}^{-2}$	31. $A_{12}^{+2}\ B_3^{+1}\ C^{+5}X_{16}^{-2}$	50. $A_9^{+1}\ B_6^{+3}\ C^{+5}X_{16}^{-2}$	
12. $A_6^{+2}\ B^{+1}C^{+3}X_8^{-2}$	32. $A_{28}^{+2}\ B_3^{+1}\ C^{+5}X_{32}^{-2}$	51. $A_{19}^{+1}\ B_{10}^{+3}\ C_3^{+5}\ X_{32}^{-2}$	68. $A_{25}^{+1}\ B_4^{+6}\ C_3^{+5}\ X_{32}^{-2}$
13. $A_{26}^{+2}\ B_3^{+1}\ C_3^{+3}\ X_{32}^{-2}$		52. $A_5^{+1}\ B_2^{+3}\ C^{+5}X_8^{-2}$	
14. $A_{14}^{+2}\ B^{+1}C^{+3}X_{16}^{-2}$	33. $A_4^{+1}\ B_3^{+2}\ C^{+6}X_8^{-2}$	53. $A_{21}^{+1}\ B_6^{+3}\ C_5^{+5}\ X_{32}^{-2}$	69. $A_{25}^{+1}\ B_5^{+5}\ C_2^{+7}\ X_{32}^{-2}$
15. $A_{30}^{+2}\ B^{+1}C^{+3}X_{32}^{-2}$	34. $A_{17}^{+2}\ B_{12}^{+1}\ C_3^{+6}\ X_{32}^{-2}$	54. $A_{11}^{+1}\ B_3^{+5}\ C_2^{+3}\ X_{16}^{-2}$	70. $A_{13}^{+1}\ B_2^{+7}\ C^{+5}X_{16}^{-2}$
	35. $A_{20}^{+1}\ B_7^{+2}\ C_5^{+6}\ X_{32}^{-2}$	55. $A_{23}^{+1}\ B_7^{+5}\ C_2^{+3}\ X_{32}^{-2}$	
16. $A_{14}^{+1}\ B_{11}^{+2}\ C_7^{+4}\ X_{32}^{-2}$	36. $A_{11}^{+2}\ B_4^{+1}\ C^{+6}X_{16}^{-2}$		71. $A_{13}^{+1}\ B_2^{+6}\ C^{+7}X_{16}^{-2}$
17. $A_7^{+2}\ B_6^{+1}\ C_3^{+4}\ X_{16}^{-2}$	37. $A_{12}^{+1}\ B_3^{+6}\ C^{+2}X_{16}^{-2}$	56. $A_{18}^{+1}\ B_{13}^{+3}\ C^{+7}X_{32}^{-2}$	
18. $A_2^{+1}\ B^{+2}C^{+4}X_4^{-2}$	38. $A_{27}^{+2}\ B_4^{+1}\ C^{+6}X_{32}^{-2}$	57. $A_{10}^{+1}\ B_5^{+3}\ C^{+7}X_{16}^{-2}$	

was developed, which makes it possible to enumerate some of ordering versions.

This program is based on the algorithm that was previously used to analyze CF4. In the first stage, all stoichiometries were sought after with allowance for the global valence balance. As a result, 2 two-component (Table 8), 71 three-component (Table 8), 387 four-component, 357 five-component, 4 six-component, and 44 seven-component stoichiometries were found. In the second stage, all versions of cation sublattice occupation (charges from +1 to +7) were enumerated for each stoichiometry, with limitation implied by the local valence balance, because it is rather difficult to consider all ordering versions, the number of which is no less than 865 (the sum of all stoichiometries). In the third and fourth stages, respectively, geometrically compatible and interpenetrating CFs were rejected.

Using this program, we obtained a number of orderings in the NaCl structure (Figs. 4–7):

I. $A^{+1}B^{+3}X_2^{-2}$ (6 orderings (Figs. 4, 5));

II. $A^{+1}B_6^{+2}C^{+3}X_8^{-2}$ (4 orderings, one tetragonal and one cubic (Fig. 6));

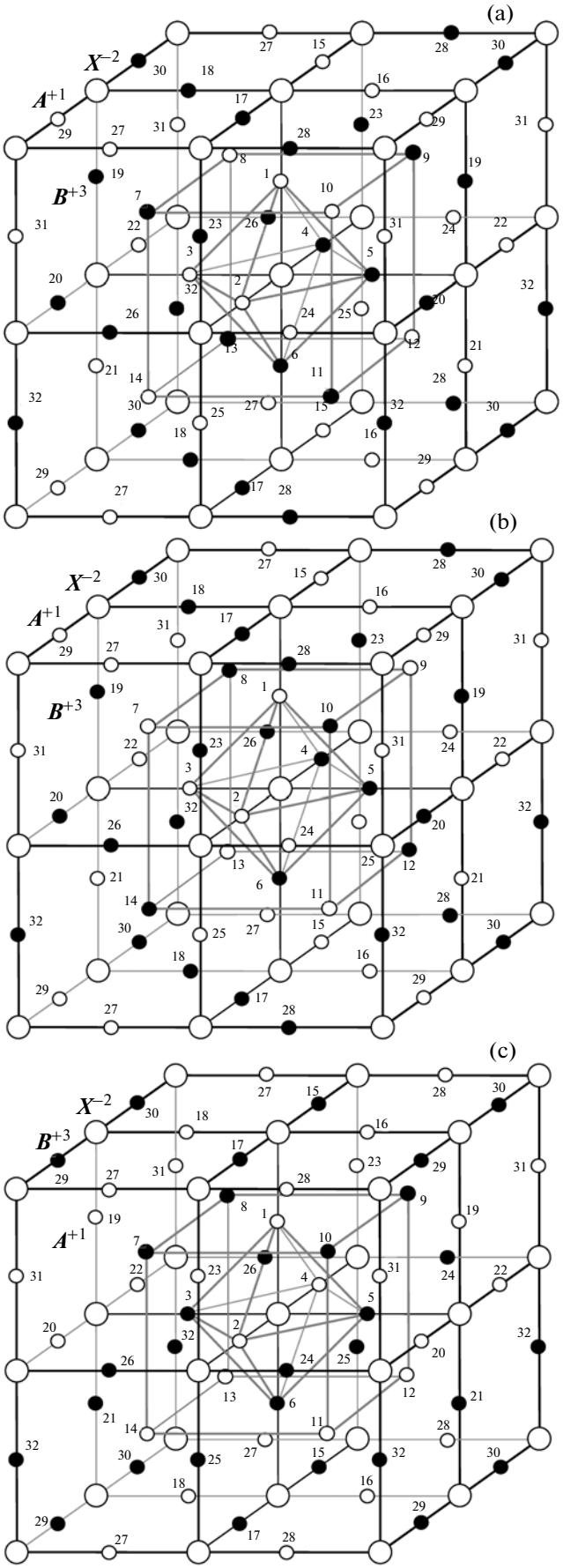

Fig. 4. Known structures corresponding to the $A^{+1}B^{+3}X_2^{-2}$ ordering for CF32: (a) rhombohedral α-$NaFeO_2$ (all octahedra are face ones), (b) cubic $LiVO_2$ (all octahedra are face ones), and (c) tetragonal γ-$LiFeO_2$ (all octahedra are edge ones).

III. $A_3^{+1}B_{10}^{+2}C_3^{+3}X_{16}^{-2}$ (3 orderings);

IV. $A^{+1}B_2^{+2}C^{+3}X_4^{-2}$ (12 orderings, one tetragonal (Fig. 7a));

V. $A_3^{+1}B_2^{+2}C_3^{+3}X_8^{-2}$ (11 orderings).

3.4. Comparison of the Results with Real Superstructures

We could visually select two unit cells from the found CF32 colorings (Fig. 7, versions I and IV).

A unit cell is selected from CF according to the Bravais rule: the cell volume must be minimum with a maximum number of symmetry elements; the valence balance (both global and local) must remain the same as in CF. The $A^{+1}B^{+3}X_2^{-2}$ structure completely coincides with the γ-$LiFeO_2$ (tetragonal) phase [4]. A known analog for the $A^{+1}B_2^{+2}C^{+3}X_4^{-2}$ (tetragonal) structure is absent; however, there is a β-Li_2CuZrO_4 phase [8] of the same structure, with a charge of +4 instead of +3, and the sums of charges on oxygen atoms differ: 13 on one of them and 11 on the other.

In Table 9 the predicted superstructures are compared with the known compounds. One can see that most compounds with the stoichiometries listed in Table 8 have not been synthesized. A more detailed analysis revealed three known orderings of the $A^{+1}B^{+3}X_2^{-2}$ type (Fig. 4); the others are unknown (Fig. 5).

Let us consider the structures obtained from CF4 taking into account only the global valence balance.

I. $A^{+5}B_3^{+1}X_4^{-2}$ (sums of charges on four octahedra: 6, 14, 14, 14). This ordering has not been found in real structures; it is likely to require a larger unit cell, for example, the Li_3NbO_4 (sums of charges on octahedra: 18, 10, 10, 10) or Li_3UO_4 (sums of charges on octahedra: 14, 14, 10, 10) structures.

II. $A^{+1}B^{+3}X_2^{-2}$ (sums of charges on octahedra: 10, 10, 14, 14). This ordering corresponds to the β-$LiFeO_2$ structure [9].

III. $A_2^{+1}B^{+2}C^{+4}X_4^{-2}$(sums of charges on octahedra: 8, 12, 14, 14). This ordering has not been observed in real structures. It is likely that CF4 is small for this stoichiometry, and the latter prefers the β-Li_2CuZrO_4 structure [8] (sums of charges on octahedra: 13, 11) because of the more uniform charge distribution.

IV. $A_2^{+2}B^{+1}C^{+3}X_4^{-2}$ (sums of charges on octahedra: 10, 14, 12, 12). This ordering has not been found in real structures. CF4 is likely to be small for this sto-

ichiometry, and the $A^{+1}B_2^{+2}C^{+3}X_4^{-2}$ structure (sum of charges on octahedra: 12, CF32) corresponds to it to a greater extent due to the more uniform charge distribution.

It should be noted that all these considerations hold true for antistructures; for example, γ-$LiFeO_2$ can be put into correspondence with Sr_2NF [11].

4. ANALYSIS OF CF4 TAKING INTO ACCOUNT INCOMPLETE OCCUPATION OF CATION AND/OR ANION SUBLATTICE IN A NaCl-TYPE STRUCTURE

If there is an incomplete occupation of the cation sublattice, the stoichiometry can be found in the same way as in Subsection 3.2. In [1], these structures were defined as geometrically related to the NaCl type (for example, $CdCl_2$, $Cu_2(OH)_3Cl$, and TiO_2 (anatase)). In this consideration it is convenient to denote the absent cation position as "□" (zero positional charge) and enumerate the charges of each element (and the empty position □) not in the septenary number system but in the octal system.

Let us consider relation II from Subsection 3.2. At a stoichiometry of 1 : 1, the sum of cation charges is $2 \times (a + b) = 8$, where the a and b charges can take values from 0 to +7, and $a \neq b$. It is convenient to enumerate all a and b in the octal system from 10 to 70 (the first and second decimals are a and b, respectively, and either a or b is zero). The global valence balance is satisfied only for the charges $a = 4$ and $b = 0$, or $A_2^{+4}\square_2 X_4^{-2}$, or $A^{+4}O_2$ (TiO_2 [20]).

The stoichiometries for relations I, III, and IV from Subsection 3.2 are as follows:

I. Absent;

III. $\square_2A^{+1}B^{+7}X_4^{-2}$ or $A^{+1}B^{+7}O_4$, $\square_2A^{+2}B^{+6}X_4^{-2}$ or $A^{+2}B^{+6}O_4$, $\square_2A^{+3}B^{+5}X_4^{-2}$ or $A^{+3}B^{+5}O_4$, $A_2^{+1}\square B^{+6}X_4^{-2}$ or $A_2^{+1}\square B^{+6}O_4$, $A_2^{+2}\square B^{+4}X_4^{-2}$ or $A_2^{+2}\square B^{+4}O_4$, $A_2^{+3}\square B^{+2}X_4^{-2}$ or $A_2^{+3}B^{+2}O_4$;

IV. $\square A^{+1}B^{+2}C^{+5}X_4^{-2}$ or $A^{+1}B^{+2}C^{+5}O_4$, $\square A^{+1}B^{+3}C^{+4}X_4^{-2}$ or $A^{+1}B^{+3}C^{+4}O_4$.

If there is an incomplete occupation of the anion sublattice, the stoichiometries can be found in the same way as in Subsection 3.2, with the following difference: the sum of cation charges at certain ratios (1 : 3, 1 : 1, 2 : 1: 1, and 1 : 1: 1: 1) will be 6, 4, and 2

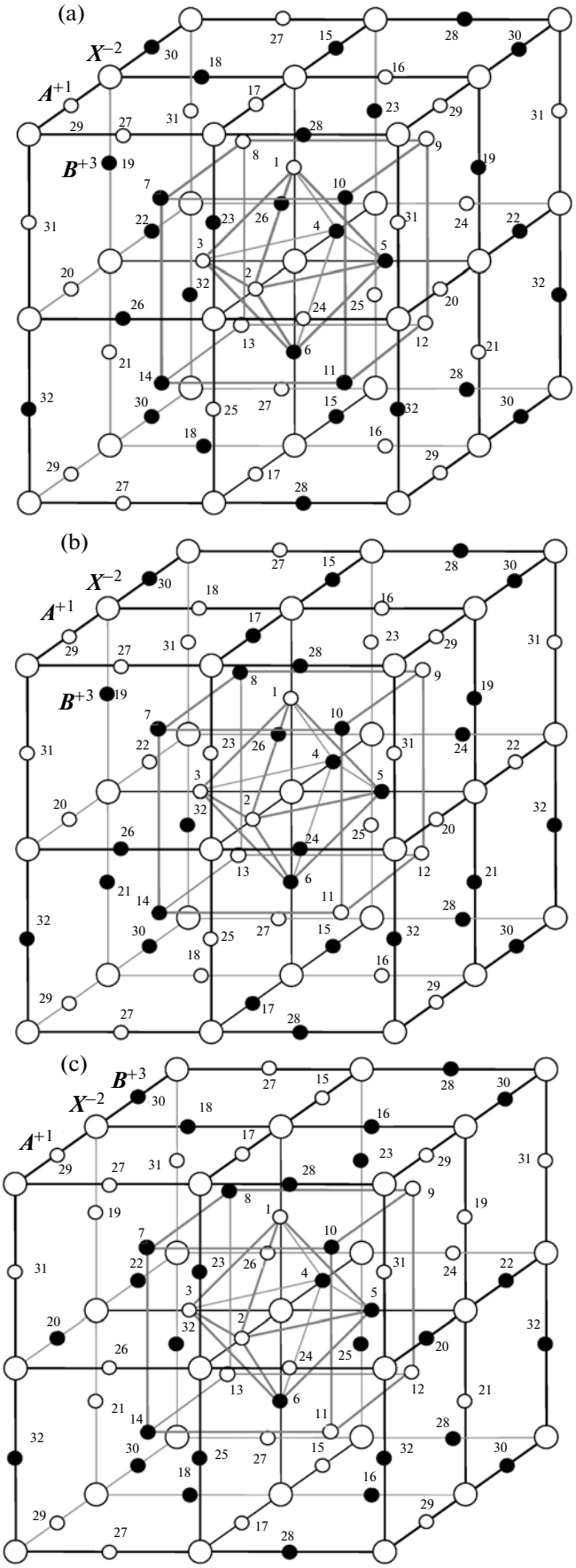

Fig. 5. Unknown orderings for CF32: (a) no. 1 (16 face and 16 edge octahedra) with a mirror plane (passes through positions 4, 7, and 11), (b) no. 2 (16 face and 16 edge octahedra) with a mirror plane (passes through positions 4, 7, and 11), and (c) no. 3 (8 face and 24 edge octahedra) with a threefold axis (passes through positions 7 and 12).

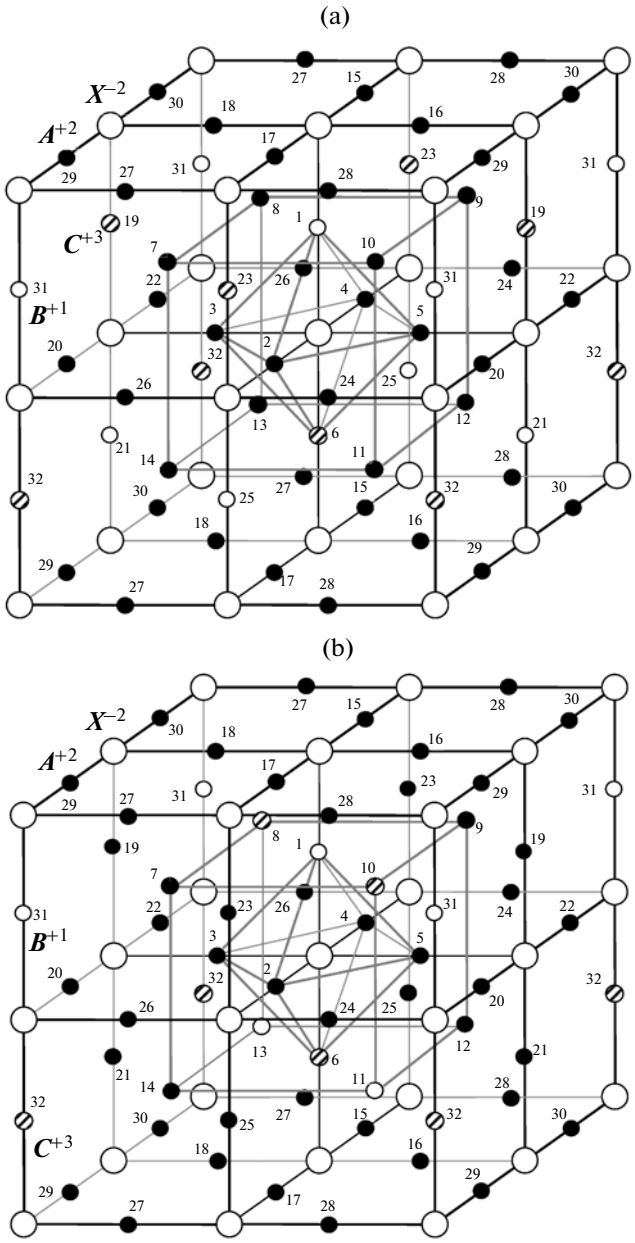

Fig. 6. (a) Tetragonal and (b) cubic $A^{+1}B_6^{+2}C^{+3}X_8^{-2}$ orderings for CF32.

(respectively, one, two, and three anions are absent). We found $A_3^{+1}B^{+3}O_3$ and $((A^{+1}B^{+2})_2O_3$ stoichiometries.

A simultaneous consideration of the unoccupied cation and anion positions leads to the following possible stoichiometries: $A^{+6}O_3$ (ReO_3 [21], UO_3 [22]), $A_2^{+3}O_3$, $A^{+1}B^{+5}O_3$, $A^{+2}B^{+4}O_3$, $A_2^{+1}B^{+4}O_3$, $A_2^{+1}B^{+2}O_2$, and $A^{+1}B^{+2}C^{+3}O_3$.

We will not report a full review of these stoichiometries, because the corresponding structures are only to be simulated.

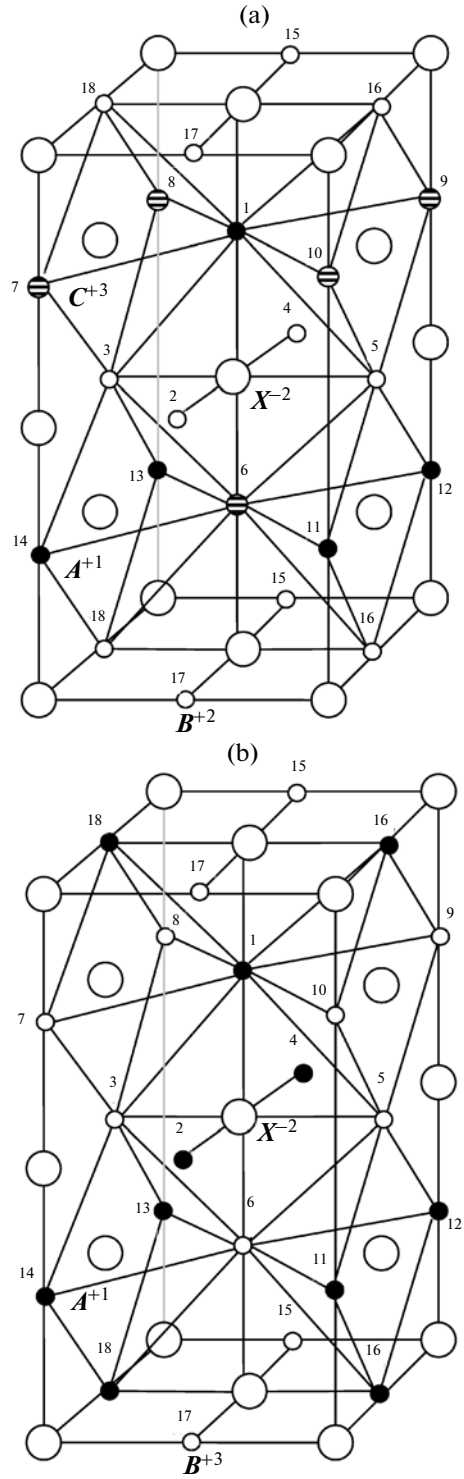

Fig. 7. Selection of smaller cells in CF32: (a) $A^{+1}B_2^{+2}C^{+3}X_4^{-2}$ and (b) $A^{+1}B^{+3}X_2^{-2}$ orderings.

5. DIFFICULTIES IN COMPARING THE MODELS OF SUPERSTRUCTURES WITH REAL ONES

(1) The CF size is not large enough to describe many structures.

Table 9. Known superstructures of the NaCl type

CF	Stoichiometry	System	Space group	Compound	Local balance valence (yes/no)	Figure	References
CF4	$A^{+1}B^{+3}X_2^{-2}$	cubic	$Fm\bar{3}m$	β-$LiFeO_2$	no	3b	[9]
CF32	$A^{+1}B^{+3}X_2^{-2}$	rhombohedral	$R\bar{3}m$	α-$NaFeO_2$	yes	4a	[4]
		cubic	$Fd\bar{3}m$	$LiVO_2$	yes	4b	[10]
		tetragonal	$I4_1/amd$	γ-$LiFeO_2$	yes	4c	[4]
	$A^{+2}{}_2X^{-3}Y^{-1}$	cubic	$Fd\bar{3}m$	Sr_2NF	yes		[11]
	$A_3^{+1}B^{+5}X_4^{-2}$	cubic	$I\bar{4}3m$	Li_3NbO_4	no		[12]
		tetragonal	$I4/mmm$	Li_3UO_4	no		[13]
		tetragonal	$P4_2/mnm$	Li_3BiO_4	no		[14]
		tetragonal	$P4_122$	Ag_3RuO_4	no		[15]
		monoclinic	$P12/a1$	Li_3RuO_4	no		[16]
		monoclinic	$C12/c1$	Li3TaO_4	no		[17]
		monoclinic	$P12/c1$	K_3SbO_4	no		[18]
		monoclinic	$P12/c1$	Li_3SbO4	no		[19]
	$A_2^{+1}B^{+2}C^{+4}X_4^{-2}$	tetragonal	$I\bar{4}m2$	β-Li_2CuZrO_4	no		[8]

(2) Real structures undergo significant distortions, especially when a structure contains unoccupied ion positions.

(3) Identical charges can be of different types.

CONCLUSIONS

The approach proposed here is general to a great extent and can be used in the structural derivation of NaCl-type superstructures. This simulation is clear and universal; in addition, different sample conditions, CF sizes, stoichiometries, topologies of anion coordination, symmetries, valence balances, and partial colorings can be set. By adding one more color, one can take into account the deficit of cations and/or anions.

The impossibility of forming superstructures with four FUs (with the complete occupation of ion positions and local valence balance) was demonstrated. CF32 with two or more colors was analyzed with allowance for the local valence balance, and several possible versions of superstructure design were found. When the octahedral environment of anions is known, one can qualitatively estimate the energetic stability of a particular ordering version.

Cations and/or anions for designed ordering models can be chosen both theoretically (minimizing the interatomic interaction energy [7]) and experimentally. Some new compounds of the $A^{+1}B^{+3}X_2^{-2}$ composition were predicted in [23] with allowance for the ratio of cation radii (r_A/r_X) and the presence of known "neighboring" structures.

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